

Fibre Optics

Fibre Optics is a technology that uses glass, plastic, threads or fibres to transmit data.

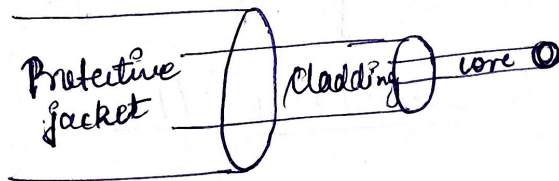
A single Optical Fibre: has its part as core, cladding, and Sheath (PROTECTING LAYER).

(Light Guiding Region)

CORE :- thin glass centre of the fibre when light travel.

CLADDING :- outer optical material surrounding the core reflect the light back into the core because cladding has lower refractive index.

SHEATH :- plastic coating that protects the fibre from damage and moisture.



BAND-WIDTH :- Difference between the upper and lower cutoff frequencies of a filter, a communication channel, or a single spectrum.

→ Typically measured in Hertz.

Fibre optics has many advantages compared with traditional metal communication :-

- i) Fibre optics cable can carry more data as their bandwidth is greater than metal cables.

$$= (A_1)^2 + A_2^2 \cos^2 \theta + 2A_1 A_2 \cos \theta + A_2^2 \sin^2 \theta$$

- (i) Fibre optics cable are less susceptible than metal cables to interference.
- (ii) Fibre optics cable are much thinner and lighter than metal wire.
- (iii) Through Fibre Optics cables the data can be transmitted digitally rather than analogically.
- (iv) Attenuation through fibre optics cables is very low in transmitting the data over a long distance so there is no need of repeater.

Fundamental Ideas of OPTICAL FIBRE :-

PYARI
MUMMY

OPTICAL FIBRE use light to carry digital signals

The base of this technology is CONCEPT OF TIR

Main concept of physics that are involved in optical fibre :-

- (i) Refraction
- (ii) Refractive indices
- (iii) Critical Angle : The incident angle at which the angle of refraction is equal to 90° is called C.A.
- (iv) Total Internal Reflection

Optical Fibre as a Dielectric Wave Guide :-

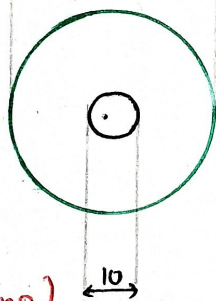
TYPES OF OPTICAL FIBRE :-

Number of modes $= \frac{V^2}{2}$

$V = \frac{2\pi a \text{NA}}{\lambda}$

SINGLE MODE
Step Index Fibre

Size: 10 μm



• Also called Single (mono) mode step index fibre

• This fibre allow single mode to propagate at a time due to very small diameter of its core.

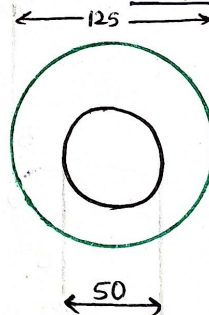
• Refractive Indices of the cladding and the core remain CONSTANT.

• Have lower single loss.

• Single mode fibres considered to be loss fibres $\rightarrow$ which increase system bandwidth & length, so these fibres are most useful for large bandwidth application.

MULTIMODE FIBRES

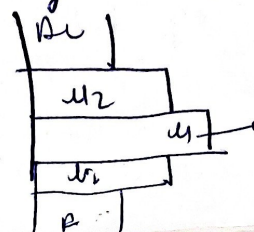
(MMF)



- Allow more than one mode to propagate.
- Size: 50 μm .

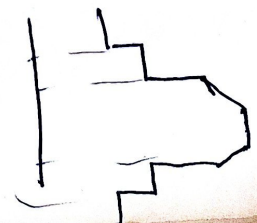
Multimode Step Index Fibres

- $\rightarrow$ Larger diameter
- $\rightarrow$ Larger NA
- $\rightarrow$ Easy to manufacture
- $\rightarrow$ Higher dispersion



Multimode Graded Index fibres

- $\rightarrow$ Concentric layers of diff RI
- $\rightarrow$ NA $\downarrow$
- $\rightarrow$ Manuf. is more complex
- $\rightarrow$ lower dispersion

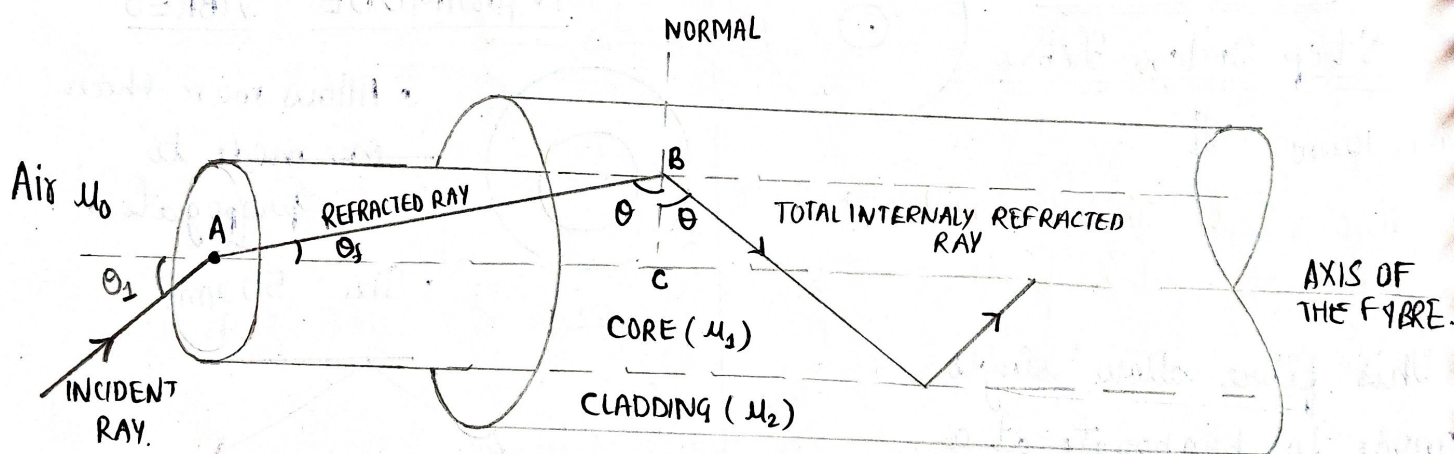


Q.10. Acceptance Angle and Numerical Aperture

In order to propagate or transmit the light wave through the optical fibre, it is necessary to launch the light at an angle that falls within a certain range.

Acceptance Angle :- Maximum angle of a ray hitting the fibre core which allows the incident light to be guided by the core.

(Maximum angle at which property of TIR occurs in fibre optics).



Applying Snell's Law at launching face of fibre :-

$$\mu_0 \sin(\theta_1) = \mu_1 \sin(\theta_f)$$

$$\frac{\sin(\theta_1)}{\sin(\theta_f)} = \frac{\mu_1}{\mu_0}$$

In ΔABC , $\theta_f + \theta + \angle C = 180^\circ$
 $\angle C = 90^\circ$

$$\theta_f + \theta = 90^\circ$$

$$\boxed{\theta_f = 90^\circ - \theta}$$

$$\frac{\sin(\theta_1)}{\sin(90-\theta)} = \frac{\mu_1}{\mu_0}$$

$$\frac{\sin(\theta_1)}{\cos(\theta)} = \frac{\mu_1}{\mu_0}$$

when $\theta = \theta_c \Rightarrow \sin \theta_1 = \frac{\mu_1}{\mu_0} \cos(\theta_c)$

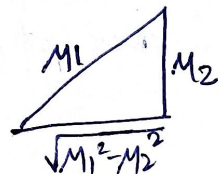
$$\frac{\sin \theta_1}{\cos \theta_c} = \frac{\mu_1}{\mu_0} \quad \text{--- (i)}$$

Snell's law at B

$$\mu_1 \sin \theta_c = \mu_2 \sin 90^\circ$$

$$\frac{\mu_1}{\mu_2} = \frac{\sin 90}{\sin \theta_c}$$

$$\sin \theta_c = \frac{\mu_2}{\mu_1}$$



$$\cos \theta_c = \frac{\sqrt{\mu_1^2 - \mu_2^2}}{\mu_1} \quad \text{--- (ii)}$$

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Substituting the value of $\cos \theta_c$ in eq (i)

$$\frac{\sin \theta_1}{\frac{\sqrt{\mu_1^2 - \mu_2^2}}{\mu_1}} = \frac{\mu_1}{\mu_0}$$

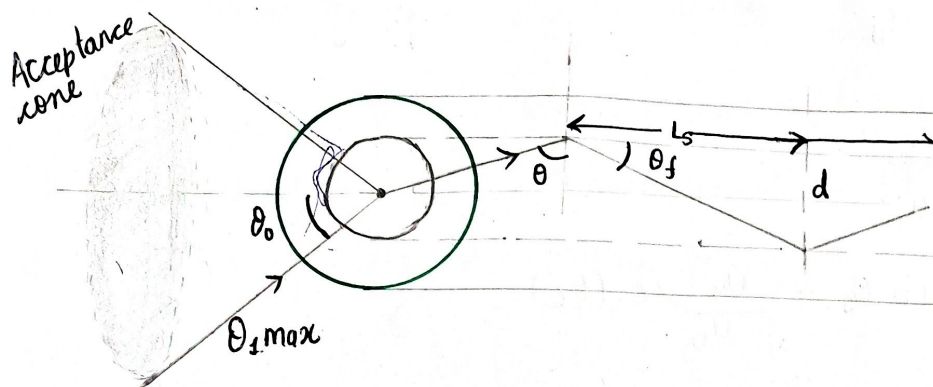
$$\sin \theta_1 = \frac{\sqrt{\mu_1^2 - \mu_2^2}}{\mu_0} = \frac{\sqrt{\mu_1^2 - \mu_2^2}}{1}$$

$$\sin \theta_{\max} = \sqrt{\mu_2^2 - \mu_1^2}$$

$$\theta_{\max} = \sin^{-1} \sqrt{\mu_2^2 - \mu_1^2}$$

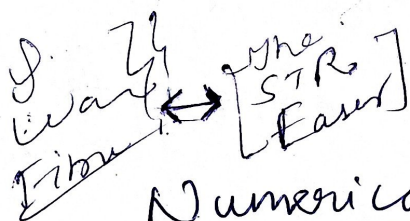
Acceptance Angle

Numerical



Acceptance cone :-

All the light rays contained in the cone having vertex angle $2\theta_{max}$ are accepted & transmitted (propagated) along the fibre. This cone is called Acceptance cone.



Numerical aperture

It is a measure of the amount of light accepted by fibre. It is defined as the sine of acceptance angle $\mu_{clad} < \mu_{core}$.

$$NA = \sin \theta_{max} = \sqrt{\mu_1^2 - \mu_2^2}$$

- 1) The refractive index for core and cladding for a step index fibre are 1.52 and 1.41 respectively. Calculate
- critical angle
 - numerical aperture
 - the maximum incidence angle.

Solⁿ:- $\mu_{\text{core}} = \mu_1 = 1.52$
 $\mu_{\text{clad}} = \mu_2 = 1.41$

$$\text{Numerical aperture} = \sqrt{\mu_1^2 - \mu_2^2}$$
$$= \sqrt{(1.52)^2 - (1.41)^2} = 0.5677 = 0.568$$

$$\text{Critical Angle} \Rightarrow \theta_c = \sin^{-1} \left[\frac{\mu_2}{\mu_1} \right] = \sin^{-1} \left[\frac{1.41}{1.51} \right] = 68.06^\circ$$

$$\text{(iii)} \quad \theta_o = \sin^{-1} \sqrt{\mu_1^2 - \mu_2^2} = \sin^{-1} \sqrt{(1.52)^2 - (1.41)^2} = \sin^{-1}(0.568)$$
$$= 34.59$$

Q₂ Find out numerical aperture and acceptance angle of an optical fibre, if the refractive index for core and cladding are 1.6 and 1.5 respectively.

Solⁿ $\mu_{\text{core}} = \mu_1 = 1.6$
 $\mu_{\text{clad}} = \mu_2 = 1.5$

$$N.A = \sqrt{\mu_1^2 - \mu_2^2} = \sqrt{(1.6)^2 - (1.5)^2} = \sqrt{0.31} = 0.556$$

$$\text{Acceptance Angle} = \theta_o = \sin^{-1} \left[\sqrt{\mu_1^2 - \mu_2^2} \right] =$$
$$= \sin^{-1}(0.556)$$
$$= 33.78^\circ$$

Ques: Calculate the numerical aperture and acceptance angle of optical fibre of refractive index for core and cladding 1.62 and 1.52 respectively.

Sol^m $\mu_{\text{core}} = \mu_1 = 1.62$

$$\mu_{\text{clad}} = \mu_2 = 1.52$$

$$N.A = \sqrt{\mu_1^2 - \mu_2^2} = \sqrt{(1.62)^2 - (1.52)^2} = 0.56$$

$$\theta_0 = \sin^{-1}(N.A)$$

$$= \sin^{-1}(0.56)$$

$$= 34.06^\circ = 34.1^\circ$$

V_{number} = It is parameter which is used to determine the maximum possible modes of a fibre.
or
Normalisation frequency

Cut-off Parameter

$$V_{\text{parameter}} = \frac{2\pi a}{\lambda} (\text{Numerical Aperture})$$

$$= \frac{2\pi a}{\lambda} \left(\sqrt{\mu_1^2 - \mu_2^2} \right)$$

$\mu_2 = \text{cladding}$
 $\mu_1 = \text{core}$

launching wavelength

$$V > 2.404 \rightarrow \text{Multimode}$$

$$V < 2.404 \rightarrow \text{Single}$$

$$\left[\text{Number of mode} = \frac{V^2}{2} \right]$$

Losses in Fiber :-

- (1) Absorption loss :-
C. due to the presence of impurities in the fiber cable some of the residue still remain
- (2) Radiative losses :- (when fibre is curved)
- (3) Microbanding losses
Macro bending losses
- Intrinsic Absorption (electronic abs. bands in the U.V. region)
Extrinsic Abs. (out run of imp in the fibre due man)

- (3) Scattering loss :- (within the fibre when there is interaction of light with density fluctuation, scattering losses occur)
- Linear Scattering losses
Non-linear Scattering losses

- (4) Dispersion loss :- (Delay distortion)
- Modal Dispersion (Diff. modes of light travel in diff. speed)
Chromatic Dispersion (Diff. wavelengths of light travel in diff. speed)

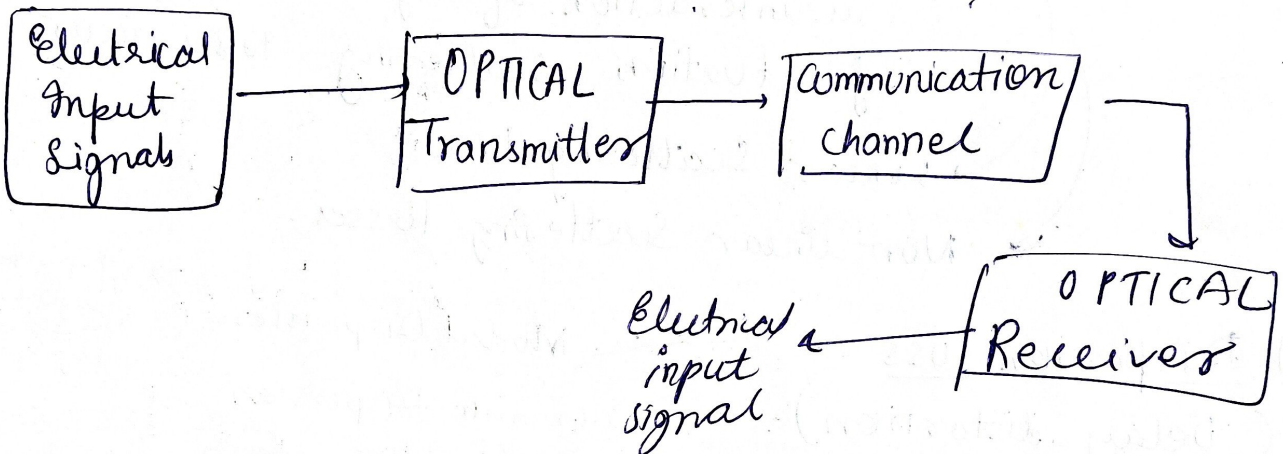
Fibre Optical Sensor :-

to measure strain, temp^r, pressure, and other quantities by modifying a fibre so that the quantity to be measured modulates the intensity, phase and polarisation.

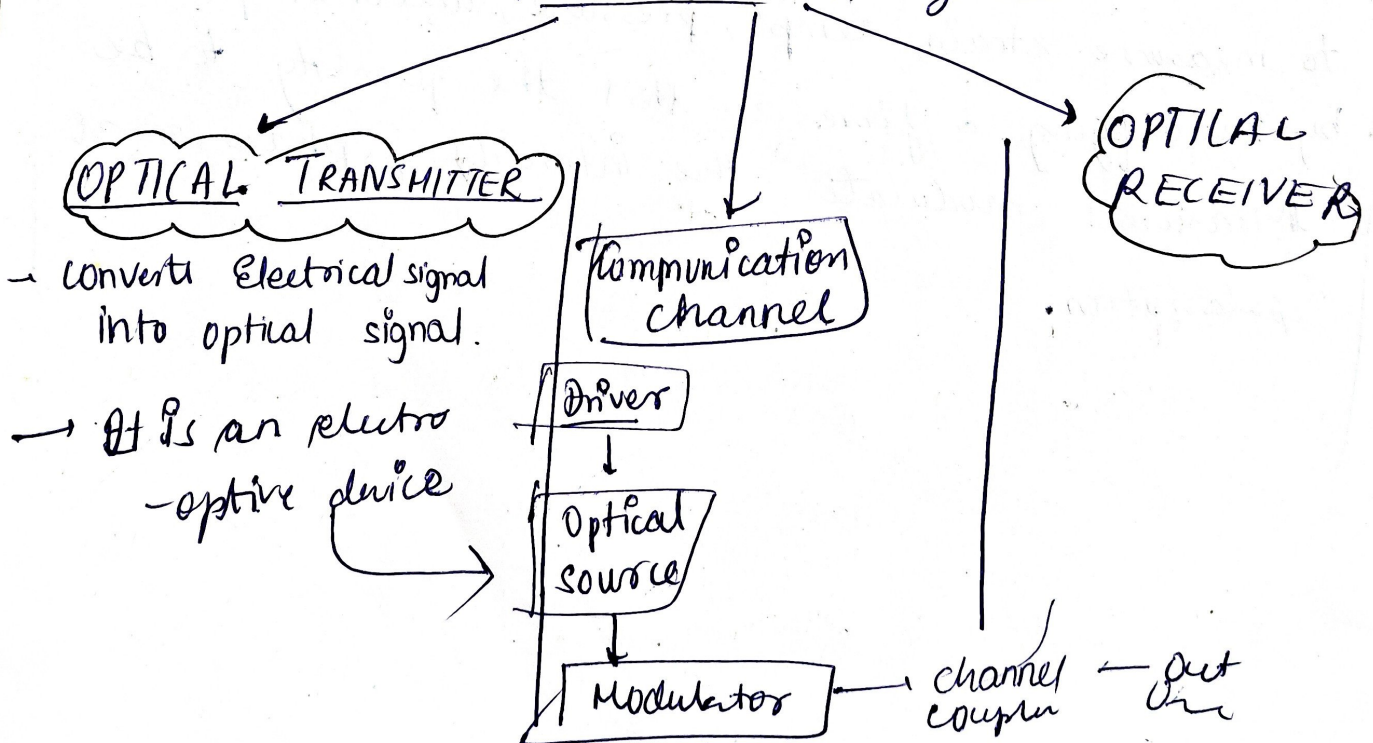
OPTICAL FIBRE COMMUNICATION SYSTEM

→ It is transmission of information by propagation of optical signals through optical fibre over an required distance.

→ can transmit voice, video through local area network computer network across long distance.



Important Components of Fibre Optics Communication System



→ Communication channel :- to transport optical signal from optical transmitter to the optical receiver without disturbing it.

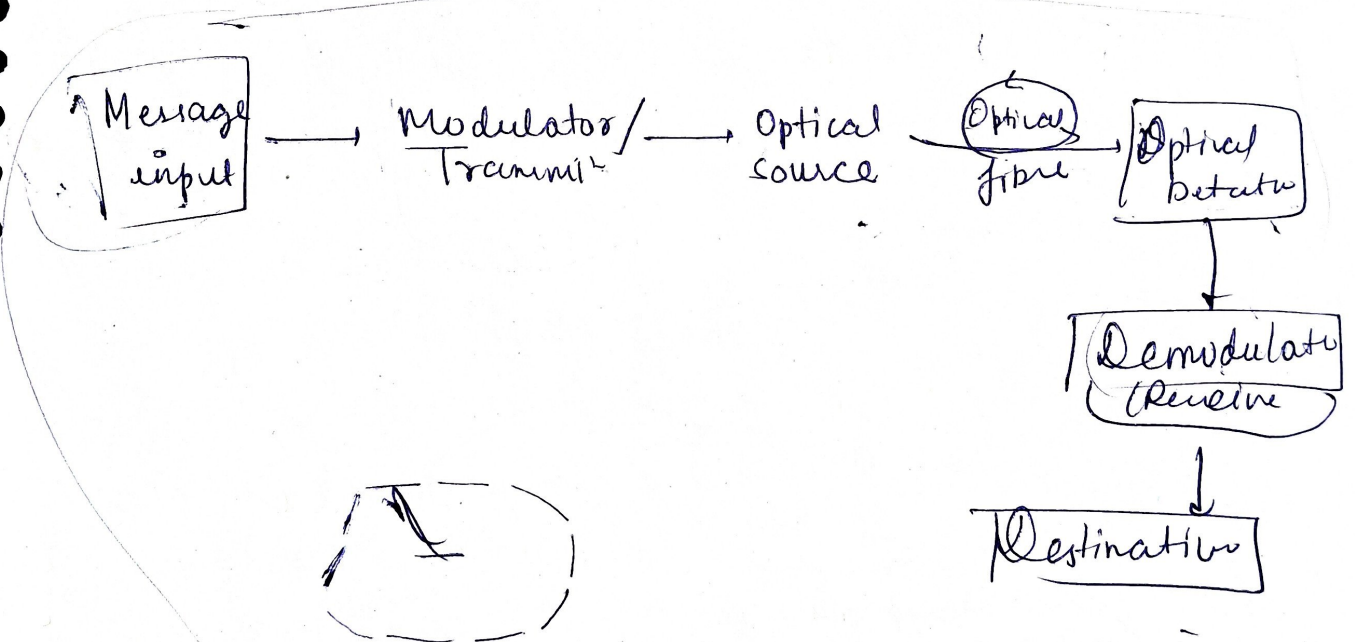
→ Optical Receiver :- It is a electro-optic device that accepts optical signal from the optical fibre and converts them into electrical signals.

Channel coupler

Photo detector

Demodulator

Ques Give the block diagram of an optical fibre communication system. Explaining the functions of different blocks compare its merits over conventional communication system.



Fibre Optical sensor :-

is a device that uses optical fibre to transmit and detect signal for sensing application.

Fibre Optical sensor are used to measure strain, pressure, temp^r and other quantities by modifying a fibre so that the quantity to be measure modulates the intensity, phase and polarisation.

Principle :-

When a light beam is sent through an optical fibre, then its parameters either in the fibre or ~~in~~ several fibre Bragg grating exp subtle changes. Then light reaches a detector, arrangement that measure these changes.

Advantages :-

- Consist of electrically insulating material
- No risk of electric spark, can be safely used in ^{explosive}
- Immune to electromagnetic interference.
- Operate over a broad range of temp^r.
- Their materials can be chemically active.

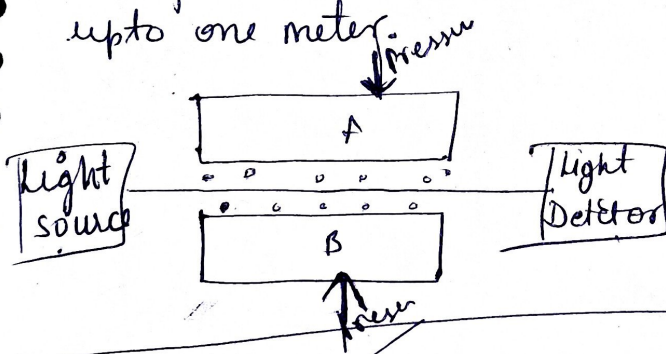


Two types :-

Intrinsic Sensor

- Sensing medium is itself a fibre. It means that the propagated light never leaves the fibre & is altered in some way by an external phenomenon,

- They can provide distributed sensing over distances of upto one meter.

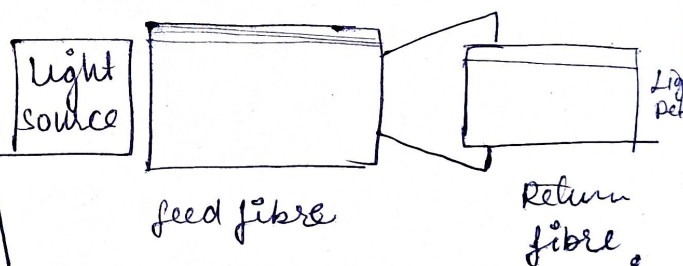


Extrinsic sensors

- Delivery of light & its collection is done by the fibre.

Thus the propagation light leaves the fibre is altered in some way & is collected by same or other fibre.

- Can reach the places which are otherwise inaccessible.



Dispersion in optical fibre

$$\lambda = nm$$

$$c = 3 \times 10^8$$

$$n_1 = 1.5$$

$$n_2 = 1.47$$

$$D = \frac{1}{c} \times \frac{1}{2} \left(\frac{n_1}{\lambda} - \frac{n_2}{\lambda} \right)$$

$$= \frac{1}{c} \times \frac{1}{2} (n_1 - n_2)$$

$$= \frac{n_1 - n_2}{2c} \Rightarrow \frac{1.5 - 1.47}{2 \times 3 \times 10^8}$$

$$D = 0.5 \times 10^{-10}$$

$$\Rightarrow \frac{0.03}{2 \times 3 \times 10^8}$$

$$\Rightarrow \frac{0.603}{2 \times 10^8} \times 10^8$$

Dispersion in optical fibre refers to the spreading of light pulses as they travel through the fibre.

Chromatic
Dispersion

Modal
Dispersion

Unit :- ~~pico second per nanometre~~
 $\frac{\text{ps}}{\text{nm-km}}$ Unit :- $\frac{\text{ps}}{\text{km}}$